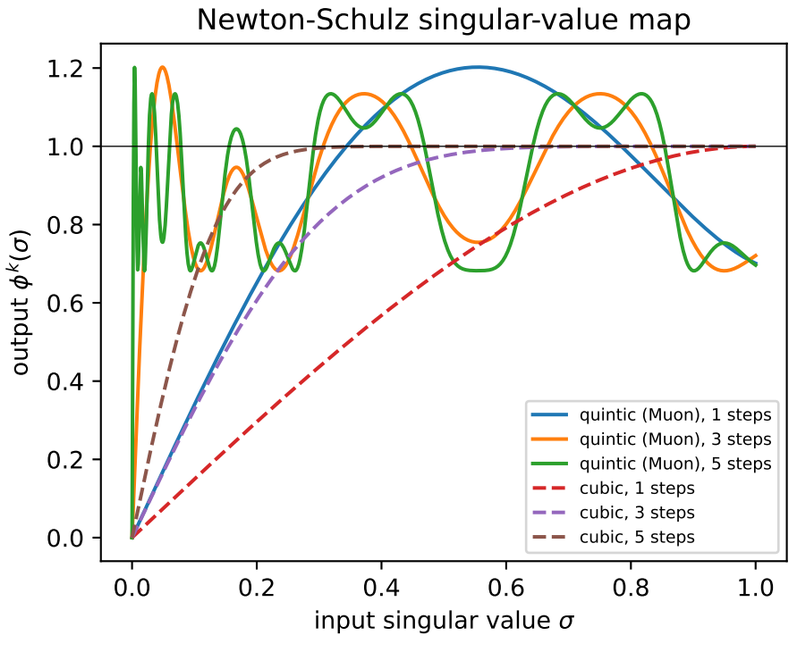
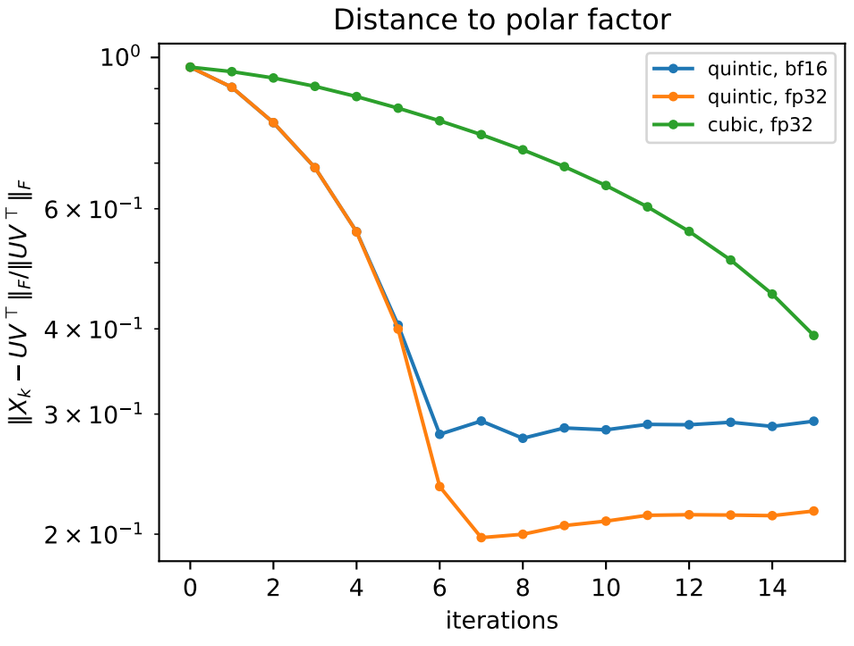
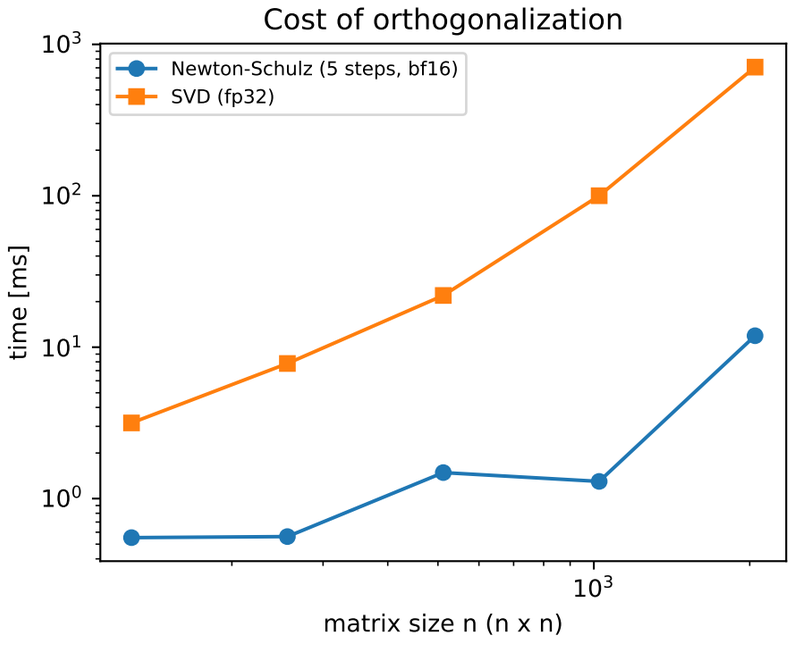

Newton-Schulz singular-value map
output $\phi^k(\sigma)$
input singular value $\sigma$
quintic (Muon), 1 steps
quintic (Muon), 3 steps
quintic (Muon), 5 steps
cubic, 1 steps
cubic, 3 steps
cubic, 5 steps


Distance to polar factor
$\|X_k - UV^\top\|_F / \|UV^\top\|_F$
iterations
quintic, bf16
quintic, fp32
cubic, fp32


Cost of orthogonalization
time [ms]
matrix size n (n x n)
Newton-Schulz (5 steps, bf16)
SVD (fp32)